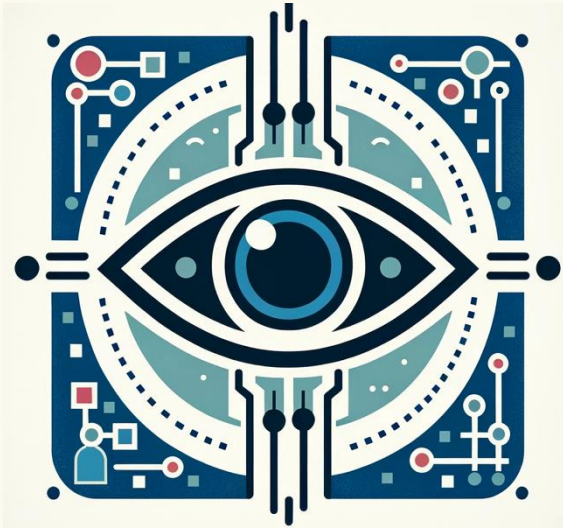
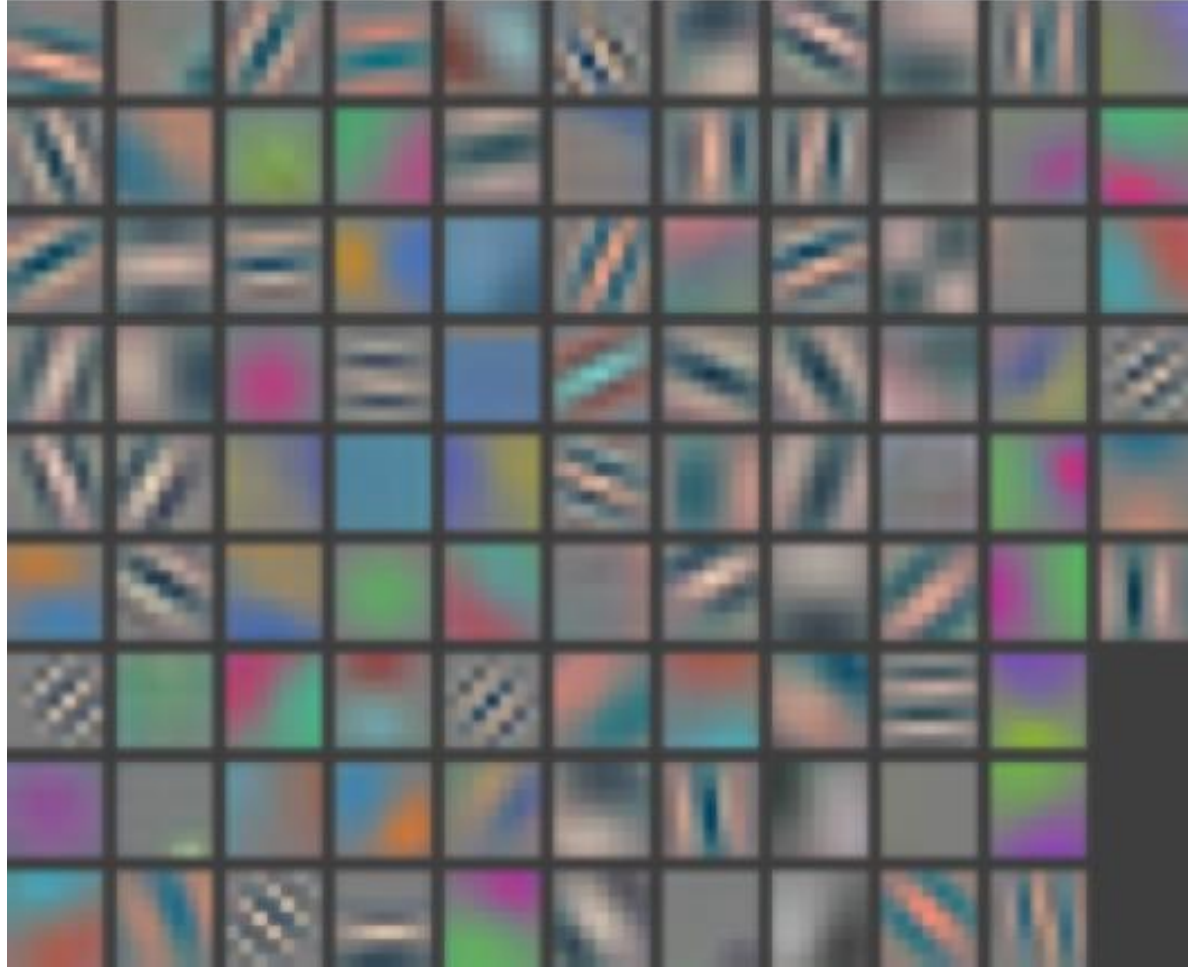




COMPUTER VISION

CNNs as Feature Extractors

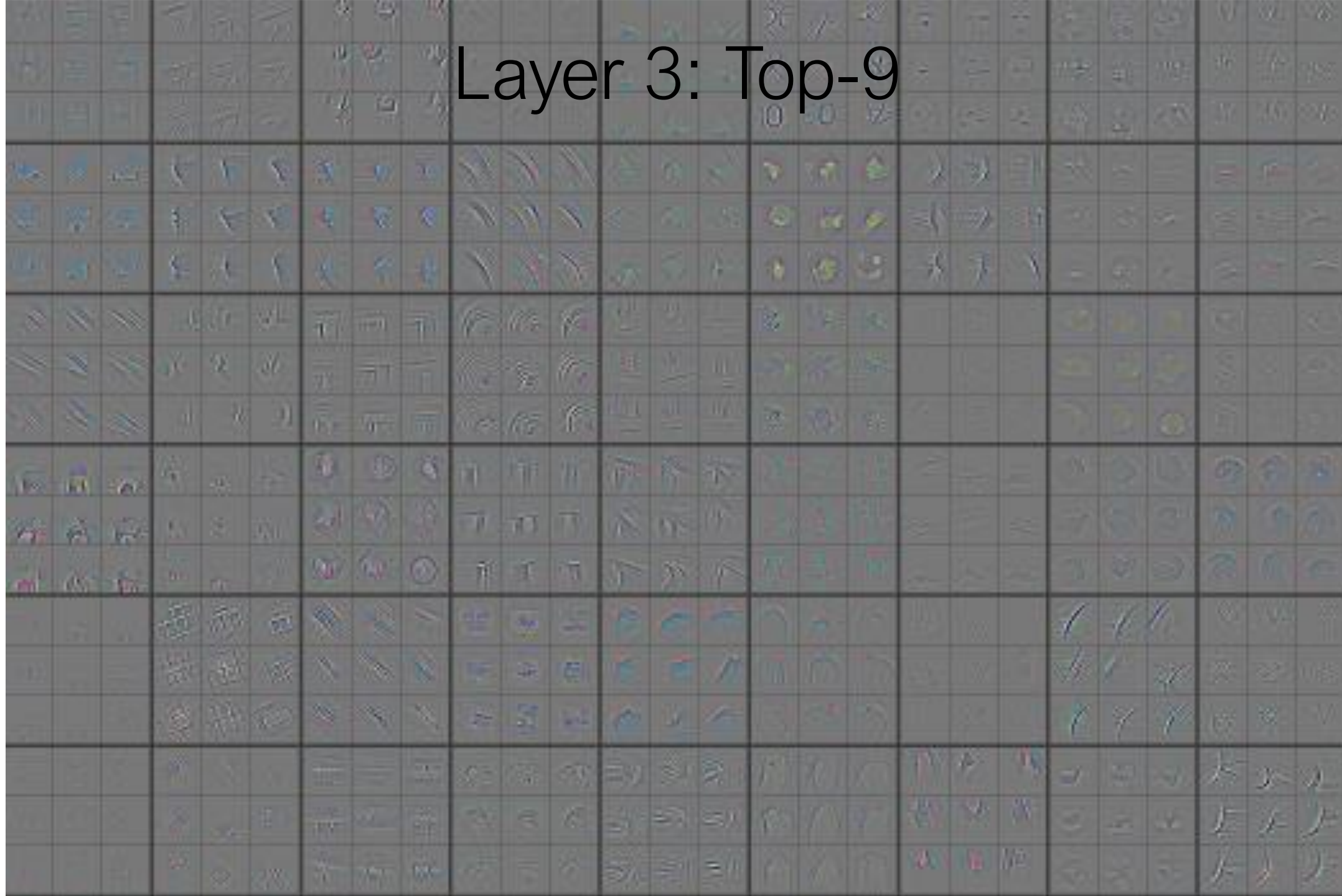
Layer 1: Filters



Layer 1: Top-9 Patches



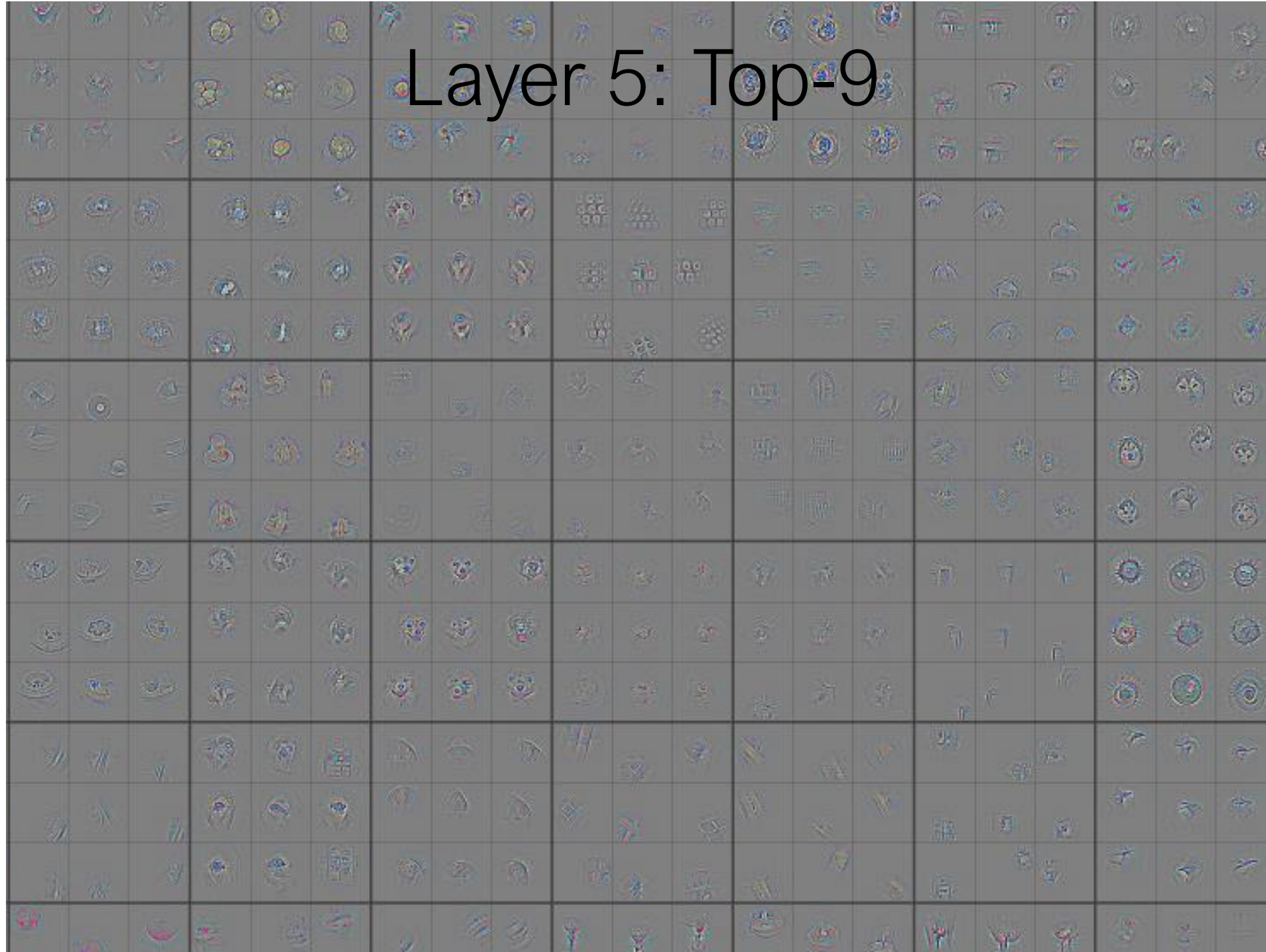
Layer 3: Top-9



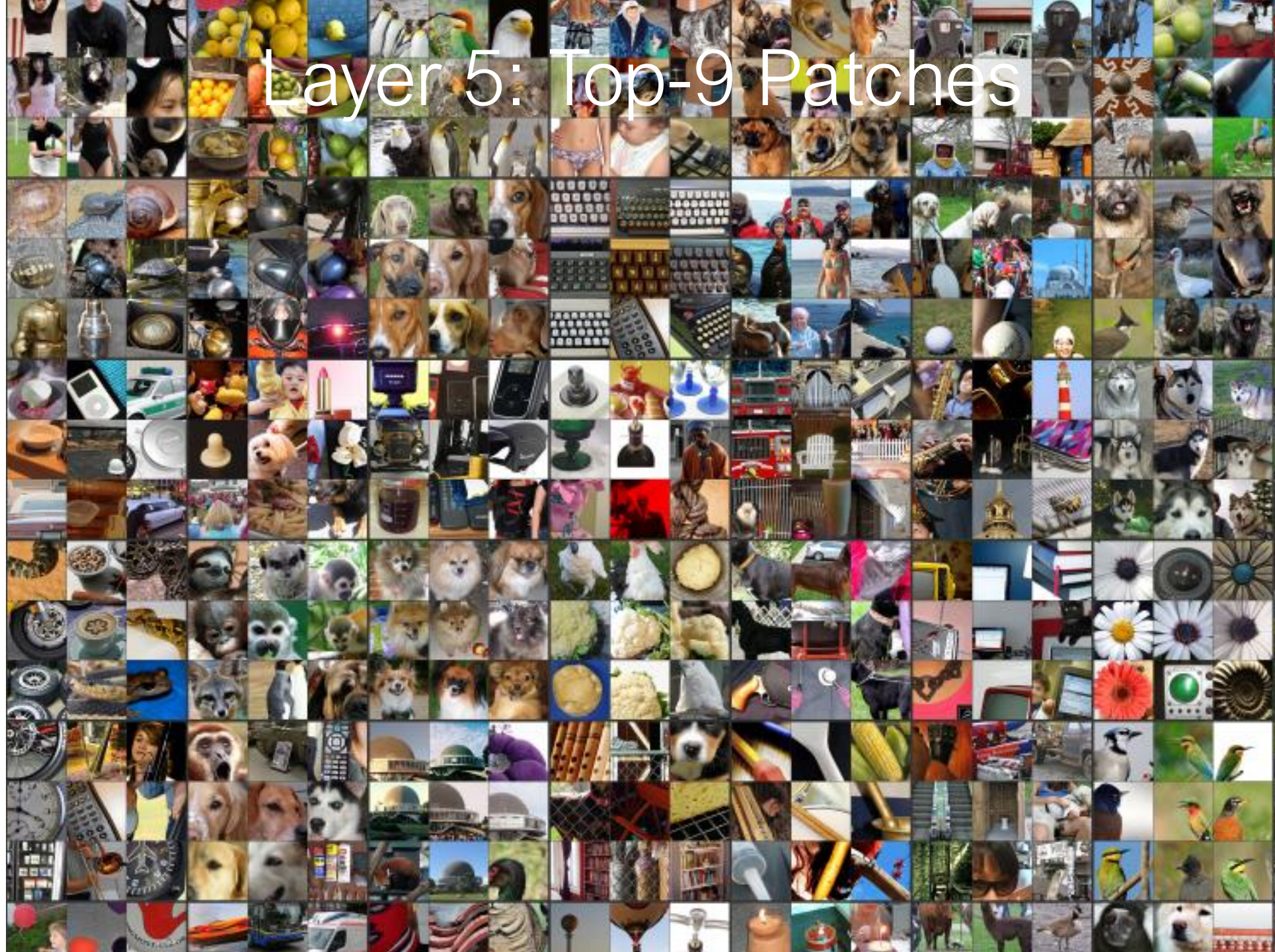
Layer 3: Top-9



Layer 5: Top-9

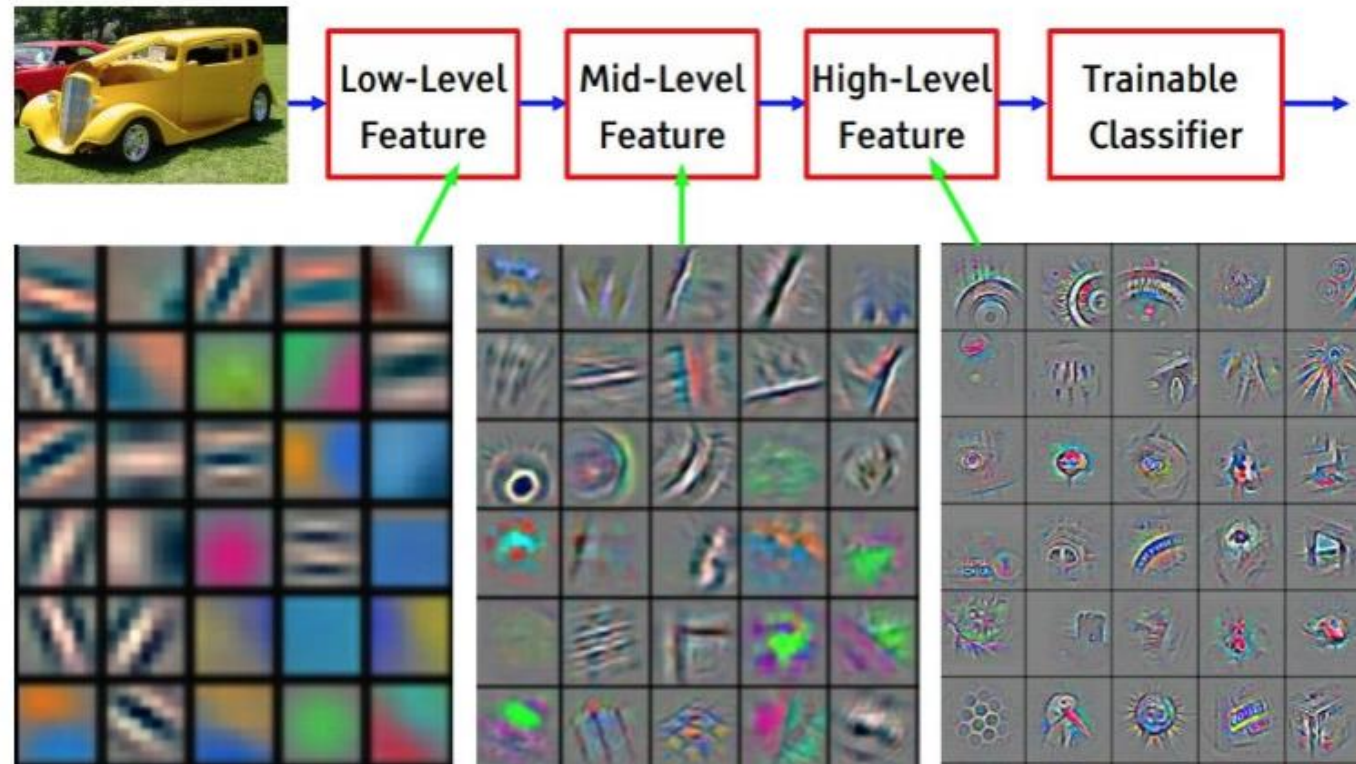


Layer 5: Top-9 Patches



Pretrained models as feature extractors

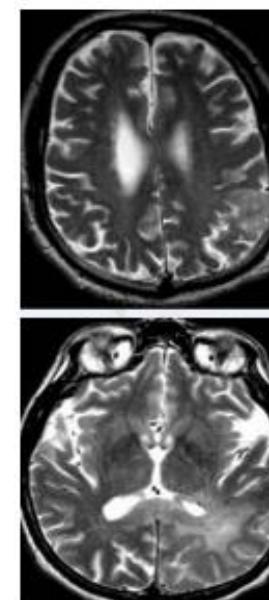
- Convolutional layers act as linear and non-linear filters able to **extract important features** for the demanded task



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]

Exercise

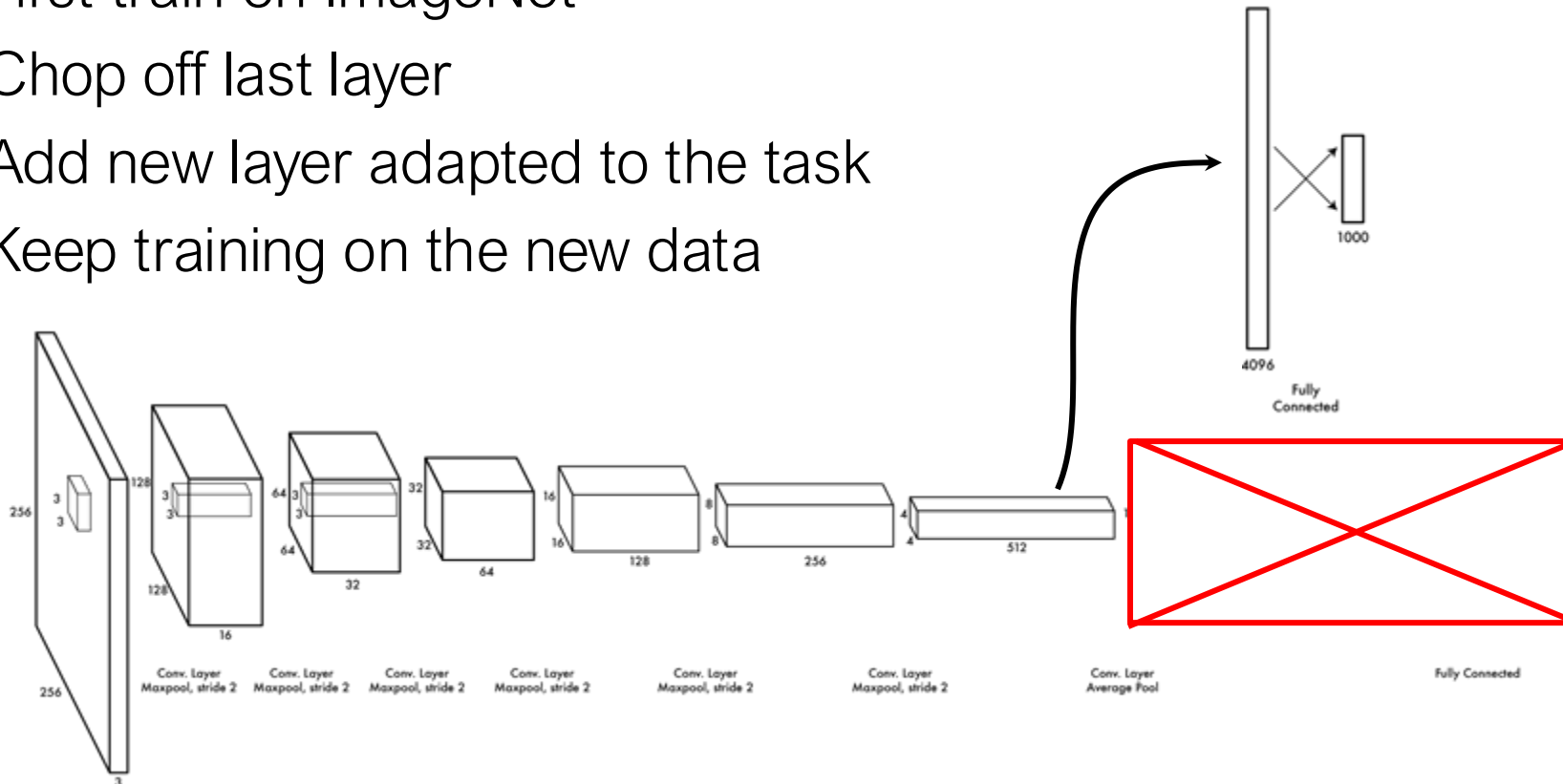
- Suppose you have the task of creating a model to classify what is the **fish species**. What strategy would you adopt to train the model?
- And a model to classify if a **brain MRI image** presents some tissue potentially with cancer



Pretraining

In fact, for dealing with smaller datasets where we might overfit, pretraining is key:

- First train on ImageNet
- Chop off last layer
- Add new layer adapted to the task
- Keep training on the new data



Pretrained models as feature extractors

- Convolutional layers as first block for another trainable computer vision problem → **transfer learning**
- Change the model slightly for a similar problem with same or different classes → **fine tuning**
- CNN trained as a vector representation of the image (**embedded vector**) and use it as input in other vision or non-vision related tasks, e.g.: captioning, translating, etc.
- Nowadays there are many **pretrained models** in all the development platforms